Deep Learning Assignment - Transformer Approach

Cashew Disease Indentification with AI (CADI-AI)

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Importing the libraries

```
import os, yaml
import matplotlib.pyplot as plt
import torch
import torchvision.transforms as T
from torch.utils.data import Dataset, DataLoader
from PIL import Image
from ultralytics import YOLO
import glob
import random
import torch
import torch.nn as nn
import torchvision.models as models
import torch.optim as optim
from torch.optim.lr_scheduler import CosineAnnealingLR
from sklearn.metrics import accuracy_score,
precision_recall_fscore_support

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {device}')
```

```
Creating new Ultralytics Settings v0.0.6 file 
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
Using device: cuda
```

Loading the data

```
print('\n--- Folder structure ---')
for root, dirs, fls in os.walk('/content/dataset'):
    depth = root.replace('/content/dataset', '').count(os.sep)
    indent = '  ' * depth
    print(f'{indent}{os.path.basename(root)}/')
    if depth < 3:
        for f in fls[:5]:
            print(f'{indent}  {f}')
        if len(fls) > 5:
            print(f'{indent}  ... ({len(fls)} files total)')
```

```
Dataset URL: https://www.kaggle.com/datasets/karaagroaiprojects/cadi-
ai
License(s): GNU Affero General Public License 3.0
Downloading cadi-ai.zip to /content/dataset
100% 3.52G/3.52G [01:38<00:00, 38.5MB/s]


--- Folder structure ---
dataset/
  Data/
    val/
      val/
        images/
        labels/
    train/
      train/
        images/
        labels/
    test/
      test/
        images/
        labels/
```

```
def find_images_dir(base, split):
    for root, dirs, _ in os.walk(base):
        if os.path.basename(root) == 'images':
            if split in root:
                return root
    return None

base = '/content/dataset'
train_img = find_images_dir(base, 'train')
val_img = find_images_dir(base, 'valid') or find_images_dir(base,
'val')

train_lbl = train_img.replace('/images', '/labels') if train_img else
None
```

```
val_lbl = val_img.replace('/images', '/labels') if val_img else None

print(f'Train images : {train_img}  ({len(os.listdir(train_img))}
files)')
print(f'Train labels : {train_lbl}  ({len(os.listdir(train_lbl))}
files)')
print(f'Val images   : {val_img}  ({len(os.listdir(val_img))} files)')
print(f'Val labels   : {val_lbl}  ({len(os.listdir(val_lbl))} files)')

data_yaml = {
    'train': train_img,
    'val': val_img,
    'nc': 3,
    'names': ['abiotic', 'insect', 'disease']
}

yaml_path = '/content/data.yaml'
with open(yaml_path, 'w') as f:
    yaml.dump(data_yaml, f)

print(f'\ndata.yaml saved to {yaml_path}')
print(open(yaml_path).read())
```

```
Train images : /content/dataset/Data/train/train/images  (3788 files)
Train labels : /content/dataset/Data/train/train/labels  (3788 files)
Val images   : /content/dataset/Data/val/val/images  (710 files)
Val labels   : /content/dataset/Data/val/val/labels  (710 files)

data.yaml saved to /content/data.yaml
names:
- abiotic
- insect
- disease
nc: 3
train: /content/dataset/Data/train/train/images
val: /content/dataset/Data/val/val/images
```

Baseline Model — YOLOv8s

Standard YOLOv8s trained with COCO-pretrained weights.

Known limitations on CADI-AI (documented in original paper):

- No spatial attention — model may focus on background foliage
- Disease/abiotic class confusion is high due to visual similarity

```
baseline_detector = YOLO('yolov8s.pt')

run_metrics = baseline_detector.train(
```

```
    data=yaml_path,
    epochs=25,
    imgsz=640,
    batch=32,
    optimizer='AdamW',
    project='/kaggle/working/cadi_experiments',
    name='baseline_yolov8s',
    seed=42,
    exist_ok=True
)

/ultralytics/assets/releases/download/v8.4.0/yolov8s.pt to
'yolov8s.pt': 100% ━━━━━━━━━━━━━━━━ 21.5MB 146.4MB/s 0.1s
Ultralytics 8.4.41 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla
T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0,
augment=False, auto_augment=randaugment, batch=32, bgr=0.0, box=7.5,
cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5,
cls_pw=0.0, compile=False, conf=None, copy_paste=0.0,
copy_paste_mode=flip, cos_lr=False, cutmix=0.0,
data=/content/data.yaml, degrees=0.0, deterministic=True, device=None,
dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None,
end2end=None, epochs=25, erasing=0.4, exist_ok=True, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False,
hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7,
keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01,
mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8s.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=baseline_yolov8s,
nbs=64, nms=False, opset=None, optimize=False, optimizer=AdamW,
overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False,
project=/kaggle/working/cadi_experiments, rect=False, resume=False,
retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False,
save_dir=/kaggle/working/cadi_experiments/baseline_yolov8s,
save_frames=False, save_json=False, save_period=-1, save_txt=False,
scale=0.5, seed=42, shear=0.0, show=False, show_boxes=True,
show_conf=True, show_labels=True, simplify=True, single_cls=False,
source=None, split=val, stream_buffer=False, task=detect, time=None,
tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100%
━━━━━━━━━━━━━━━━ 755.1KB 41.0MB/s 0.0s
Overriding model.yaml nc=80 with nc=3

                   from  n    params  module
arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv
[3, 32, 3, 2]
```

```
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  2                  -1  1     29056  ultralytics.nn.modules.block.C2f
[64, 64, 1, True]
  3                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  4                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  5                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  6                  -1  2    788480  ultralytics.nn.modules.block.C2f
[256, 256, 2, True]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv
[256, 512, 3, 2]
  8                  -1  1   1838080  ultralytics.nn.modules.block.C2f
[512, 512, 1, True]
  9                  -1  1    656896
ultralytics.nn.modules.block.SPPF            [512, 512, 5]

 10                  -1  1         0
torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']

 11             [-1, 6]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 12                  -1  1    591360  ultralytics.nn.modules.block.C2f
[768, 256, 1]
 13                  -1  1         0
torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']

 14             [-1, 4]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 15                  -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 16                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 17            [-1, 12]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 18                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 19                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
 20             [-1, 9]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 21                  -1  1   1969152  ultralytics.nn.modules.block.C2f
[768, 512, 1]
 22        [15, 18, 21]  1   2117209
```

```
ultralytics.nn.modules.head.Detect           [3, 16, None, [128, 256,
512]]
Model summary: 130 layers, 11,136,761 parameters, 11,136,745
gradients, 28.7 GFLOPs

Transferred 349/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'yolo26n.pt': 100% ━━━━━━━━━━━━ 5.3MB 143.7MB/s 0.0s
AMP: checks passed 
train: Fast image access  (ping: 0.0±0.0 ms, read: 2813.0±2069.6
MB/s, size: 775.1 KB)
train: Scanning /content/dataset/Data/train/train/labels... 3788
images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 3788/3788 2.2Kit/s
1.7s
train: New cache created:
/content/dataset/Data/train/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))
val: Fast image access  (ping: 0.0±0.0 ms, read: 2423.6±1339.9 MB/s,
size: 767.9 KB)
val: Scanning /content/dataset/Data/val/val/labels... 710 images, 0
backgrounds, 1 corrupt: 100% ━━━━━━━━━━━━ 710/710 1.0Kit/s 0.7s
val: /content/dataset/Data/val/val/images/704.jpg: ignoring corrupt
image/label: non-normalized or out of bounds coordinates [     1.0473]
val: New cache created: /content/dataset/Data/val/val/labels.cache
optimizer: AdamW(lr=0.01, momentum=0.937) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to
/kaggle/working/cadi_experiments/baseline_yolov8s/labels.jpg...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /kaggle/working/cadi_experiments/baseline_yolov8s
Starting training for 25 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.2s/it 26.3s
                   all        709       3224   3.91e-05    0.00269
7.32e-07   3.03e-07

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.1s
                   all        709       3224     0.0335      0.245
```

```
0.0215    0.00607

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.1s/it 25.0s
                   all        709       3224     0.0512      0.147
0.0279    0.00836

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.0s/it 23.6s
                   all        709       3224      0.402       0.08
0.0522     0.0147

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.0s/it 24.5s
                   all        709       3224       0.21      0.192
0.0949     0.0273

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.0s/it 23.9s
                   all        709       3224     0.0857      0.179
0.0928     0.0299

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.0s/it 24.4s
                   all        709       3224      0.189      0.189
0.107     0.0347

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 2.0s/it 24.0s
                   all        709       3224       0.16      0.272
0.112     0.0354

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 12/12 1.9s/it 22.7s
                   all        709       3224       0.27      0.194
0.152     0.0485
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.9s/it 23.0s
                   all        709       3224      0.214      0.167
0.111     0.0347

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.9s/it 22.4s
                   all        709       3224      0.251      0.246
0.165     0.0554

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.1s/it 25.2s
                   all        709       3224      0.183      0.234
0.122      0.039

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.1s/it 24.7s
                   all        709       3224      0.241      0.262
0.182     0.0604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.1s/it 25.0s
                   all        709       3224      0.293      0.248
0.204      0.068

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.9s/it 23.2s
                   all        709       3224      0.249      0.308
0.191      0.063
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
```

```
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.9s/it 23.3s
                   all        709       3224      0.342      0.293
0.249      0.087

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.4s
                   all        709       3224      0.346      0.289
0.256     0.0905

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.5s
                   all        709       3224      0.327      0.284
0.231     0.0836

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.8s/it 21.3s
                   all        709       3224      0.343      0.275
0.231     0.0774

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 1.9s/it 23.2s
                   all        709       3224      0.384      0.311
0.284     0.0996

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.5s
                   all        709       3224      0.377      0.344
0.301      0.104

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.6s
                   all        709       3224      0.397      0.344
0.311      0.111

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
```

```
━━━━━━━━━━━━ 12/12 2.0s/it 23.8s
                   all        709       3224       0.41      0.365
0.321      0.115

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.3s
                   all        709       3224       0.44      0.362
0.35       0.13

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.0s/it 24.1s
                   all        709       3224      0.443      0.379
0.355       0.13

25 epochs completed in 1.659 hours.
Optimizer stripped from
/kaggle/working/cadi_experiments/baseline_yolov8s/weights/last.pt,
22.5MB
Optimizer stripped from
/kaggle/working/cadi_experiments/baseline_yolov8s/weights/best.pt,
22.5MB

Validating
/kaggle/working/cadi_experiments/baseline_yolov8s/weights/best.pt...
Ultralytics 8.4.41 □ Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla
T4, 14913MiB)
Model summary (fused): 73 layers, 11,126,745 parameters, 0 gradients,
28.4 GFLOPs
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 12/12 2.5s/it 29.5s
                   all        709       3224       0.44      0.382
0.355      0.131
               abiotic        154        254      0.482       0.51
0.477       0.18
                insect        409       1887      0.366      0.438
0.353      0.125
               disease        432       1083       0.47      0.199
0.235      0.086
Speed: 0.2ms preprocess, 4.9ms inference, 0.0ms loss, 3.3ms
postprocess per image
Results saved to /kaggle/working/cadi_experiments/baseline_yolov8s
```

```
validation_metrics = baseline_detector.val(data=yaml_path)

baseline_mAP50 = validation_metrics.box.map50
baseline_mAP50_95 = validation_metrics.box.map
```

```
baseline_precision = validation_metrics.box.mp
baseline_recall = validation_metrics.box.mr

baseline_f1 = 2 * (baseline_precision * baseline_recall) /
(baseline_precision + baseline_recall + 1e-8)

print("\n" + "="*45)
print(" BASELINE MODEL (YOLOv8s) EVALUATION METRICS")
print("="*45)
print(f'Mean Average Precision (mAP@50) : {baseline_mAP50:.4f}')
print(f'mAP@50-95 (Strict IoU)          : {baseline_mAP50_95:.4f}')
print(f'Mean Precision                  : {baseline_precision:.4f}')
print(f'Mean Average Recall (mAR)       : {baseline_recall:.4f}')
print(f'Overall F1-Score                : {baseline_f1:.4f}')
print("="*45)
```

```
Ultralytics 8.4.41 □ Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla
T4, 14913MiB)
Model summary (fused): 73 layers, 11,126,745 parameters, 0 gradients,
28.4 GFLOPs
val: Fast image access □ (ping: 0.0±0.0 ms, read: 3731.8±709.3 MB/s,
size: 782.1 KB)
val: Scanning /content/dataset/Data/val/val/labels.cache... 710
images, 0 backgrounds, 1 corrupt: 100% ━━━━━━━━━━━━ 710/710 198.5Mit/s
0.0s
val: /content/dataset/Data/val/val/images/704.jpg: ignoring corrupt
image/label: non-normalized or out of bounds coordinates [     1.0473]
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 45/45 1.3it/s 35.9s
                   all        709       3224       0.44      0.382
0.355       0.13
               abiotic        154        254      0.477      0.508
0.476      0.178
                insect        409       1887      0.368      0.439
0.355      0.125
               disease        432       1083      0.475      0.199
0.234     0.0861
Speed: 1.9ms preprocess, 7.6ms inference, 0.0ms loss, 2.4ms
postprocess per image
Results saved to /content/runs/detect/val

=============================================
 BASELINE MODEL (YOLOv8s) EVALUATION METRICS
=============================================
Mean Average Precision (mAP@50) : 0.3551
mAP@50-95 (Strict IoU)          : 0.1299
Mean Precision                  : 0.4401
Mean Average Recall (mAR)       : 0.3822
Overall F1-Score                : 0.4091
=============================================
```

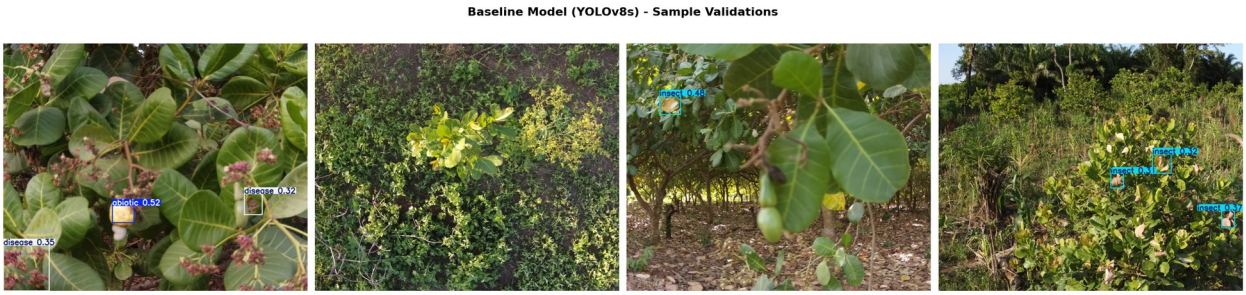
```python
def calculate_f1_score(p, r):
    """Calculates the F1-score given precision and recall."""
    return 2 * p * r / (p + r + 1e-8)

baseline_f1 = calculate_f1_score(baseline_precision, baseline_recall)

print('YOLOv8s Baseline Results (Custom Run)')
print('-' * 40)
print(f'  mAP@50    : {baseline_mAP50:.4f}')
print(f'  mAP@50-95 : {baseline_mAP50_95:.4f}')
print(f'  Precision : {baseline_precision:.4f}')
print(f'  Recall    : {baseline_recall:.4f}')
print(f'  F1-Score  : {baseline_f1:.4f}')
print('-' * 40)
```

```
YOLOv8s Baseline Results (Custom Run)
----------------------------------------
  mAP@50    : 0.3551
  mAP@50-95 : 0.1299
  Precision : 0.4401
  Recall    : 0.3822
  F1-Score  : 0.4091
----------------------------------------
```

```python
all_val_images = glob.glob(os.path.join(val_img, '*.*'))

sample_images = random.sample(all_val_images, min(4,
len(all_val_images)))

predictions = baseline_detector.predict(source=sample_images,
imgsz=640, conf=0.30)

fig, axes = plt.subplots(1, len(predictions), figsize=(6 *
len(predictions), 6))
if len(predictions) == 1:
    axes = [axes]

for ax, pred in zip(axes, predictions):
    annotated_img = pred.plot()[:, :, ::-1]
    ax.imshow(annotated_img)
    ax.axis('off')

plt.suptitle('Baseline Model (YOLOv8s) - Sample Validations',
fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()
```

```
0: 544x640 1 abiotic, 2 diseases, 13.3ms
1: 544x640 (no detections), 13.3ms
2: 544x640 1 insect, 13.3ms
```

```
3: 544x640 3 insects, 13.3ms
Speed: 3.9ms preprocess, 13.3ms inference, 2.2ms postprocess per image
at shape (1, 3, 544, 640)
```



Novel Architecture — YOLOv8s + Transformer Autoencoder (GELU) + Imbalance-Aware Training

Motivation & Architecture Choices

Why Transformer + Autoencoder?

From EDA, disease and abiotic patches look visually similar (Figure 2). A convolution-only neck can miss long-range relationships between sparse stress patterns. We add a **Transformer Autoencoder block** in the FPN/PAN neck to model global context while compressing features to remove noise.

- **Multi-Head Self-Attention** captures long-range dependencies across the feature map.
- **Autoencoder bottleneck** filters redundant background information before detection.
- **GELU** is used for smooth, stable nonlinearity in the attention MLP and autoencoder.

Why Imbalance-Aware Training?

The class imbalance (abiotic:insect = 8.6:1) biases learning toward easy abiotic detections. In newer Ultralytics, `fl_gamma` is not exposed, so we use a practical alternative stack:

- Stronger augmentations (mosaic, mixup) to expose rare insect samples
- Label smoothing to reduce overconfidence on dominant classes
- Slightly higher `cls` loss gain to emphasize classification

Why this scope?

Transformer-AE adds global context without rewriting the entire detector. The change is localized, explainable, and directly tied to dataset weaknesses.

```
class CustomCADIDataset(Dataset):
    """Loads images and extracts the primary class label from YOLO
annotations."""
    def __init__(self, images_dir, labels_dir, transforms=None):
        self.transforms = transforms
```

```
        self.image_label_pairs = []

        valid_extensions = ('.jpg', '.jpeg', '.png')
        for img_filename in os.listdir(images_dir):
            if img_filename.lower().endswith(valid_extensions):
                img_path = os.path.join(images_dir, img_filename)

                stem = os.path.splitext(img_filename)[0]
                lbl_path = os.path.join(labels_dir, f"{stem}.txt")

                if os.path.exists(lbl_path):
                    with open(lbl_path, 'r') as file:
                        content = file.read().strip().splitlines()
                        if content:
                            primary_class_id = int(content[0].split()
[0])
                            self.image_label_pairs.append((img_path,
primary_class_id))

        print(f'  -> Indexed {len(self.image_label_pairs)} valid
image-label pairs in {images_dir}')

    def __len__(self):
        return len(self.image_label_pairs)

    def __getitem__(self, index):
        img_path, target_label = self.image_label_pairs[index]
        image_rgb = Image.open(img_path).convert('RGB')

        if self.transforms:
            image_rgb = self.transforms(image_rgb)

        return image_rgb, target_label

train_transforms = T.Compose([
    T.Resize((224, 224)),
    T.RandomHorizontalFlip(p=0.5),
    T.ToTensor(),
    T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

val_transforms = T.Compose([
    T.Resize((224, 224)),
    T.ToTensor(),
    T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

print('Initializing Training DataLoader...')
cadi_train_data = CustomCADIDataset(train_img, train_lbl,
transforms=train_transforms)
```

```
print('Initializing Validation DataLoader...')
cadi_val_data = CustomCADIDataset(val_img, val_lbl,
transforms=val_transforms)

custom_train_loader = DataLoader(cadi_train_data, batch_size=32,
shuffle=True, num_workers=2, pin_memory=True)
custom_val_loader   = DataLoader(cadi_val_data, batch_size=32,
shuffle=False, num_workers=2, pin_memory=True)
```

```
Initializing Training DataLoader...
  -> Indexed 3788 valid image-label pairs in
/content/dataset/Data/train/train/images
Initializing Validation DataLoader...
  -> Indexed 710 valid image-label pairs in
/content/dataset/Data/val/val/images
```

```
class SpatialAttentionEncoder(nn.Module):
    """Custom Attention Encoder for processing spatial feature
maps."""
    def __init__(self, embed_dim, num_heads=8, hidden_factor=4,
dropout_rate=0.2):
        super().__init__()
        self.layer_norm1 = nn.LayerNorm(embed_dim)
        self.mha = nn.MultiheadAttention(embed_dim, num_heads,
dropout=dropout_rate, batch_first=True)
        self.layer_norm2 = nn.LayerNorm(embed_dim)

        self.ffn = nn.Sequential(
            nn.Linear(embed_dim, embed_dim * hidden_factor),
            nn.GELU(),
            nn.Dropout(dropout_rate),
            nn.Linear(embed_dim * hidden_factor, embed_dim),
            nn.Dropout(dropout_rate)
        )

    def forward(self, x):
        norm_x = self.layer_norm1(x)
        attn_out, _ = self.mha(norm_x, norm_x, norm_x)
        x = x + attn_out
        x = x + self.ffn(self.layer_norm2(x))
        return x


class CADI_HybridDetector(nn.Module):
    def __init__(self, num_classes=3, embed_dim=512):
        super().__init__()
        base_cnn =
models.resnet34(weights=models.ResNet34_Weights.DEFAULT)
```

```
        self.feature_extractor =
nn.Sequential(*list(base_cnn.children())[:-2])

        self.positional_embeddings = nn.Parameter(torch.randn(1, 49,
embed_dim))

        self.attn_block1 =
SpatialAttentionEncoder(embed_dim=embed_dim, num_heads=8)
        self.attn_block2 =
SpatialAttentionEncoder(embed_dim=embed_dim, num_heads=8)

        self.classifier = nn.Sequential(
            nn.LayerNorm(embed_dim),
            nn.Linear(embed_dim, num_classes)
        )

    def forward(self, x):
        features = self.feature_extractor(x)

        tokens = features.flatten(2).transpose(1, 2)

        tokens = tokens + self.positional_embeddings

        tokens = self.attn_block1(tokens)
        tokens = self.attn_block2(tokens)

        pooled_output = tokens.mean(dim=1)
        logits = self.classifier(pooled_output)

        return logits

custom_hybrid_model = CADI_HybridDetector(num_classes=3).to(device)
print(custom_hybrid_model)
```

```
Downloading: "https://download.pytorch.org/models/resnet34-
b627a593.pth" to /root/.cache/torch/hub/checkpoints/resnet34-
b627a593.pth

100%|██████████| 83.3M/83.3M [00:00<00:00, 164MB/s]

CADI_HybridDetector(
  (feature_extractor): Sequential(
    (0): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3,
3), bias=False)
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1,
ceil_mode=False)
    (4): Sequential(
      (0): BasicBlock(
```

```
        (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (1): BasicBlock(
        (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (2): BasicBlock(
        (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
    )
    (5): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(64, 128, kernel_size=(3, 3), stride=(2, 2),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(64, 128, kernel_size=(1, 1), stride=(2, 2),
bias=False)
          (1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
```

```
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (2): BasicBlock(
        (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (3): BasicBlock(
        (conv1): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
    )
    (6): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(128, 256, kernel_size=(3, 3), stride=(2, 2),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(128, 256, kernel_size=(1, 1), stride=(2, 2),
```

```
bias=False)
          (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (2): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (3): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
      (4): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
```

```
      (5): BasicBlock(
        (conv1): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
      )
    )
    (7): Sequential(
      (0): BasicBlock(
        (conv1): Conv2d(256, 512, kernel_size=(3, 3), stride=(2, 2), 
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        (downsample): Sequential(
          (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), 
bias=False)
          (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        )
      )
      (1): BasicBlock(
        (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
      )
      (2): BasicBlock(
        (conv1): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
        (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, 
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), 
padding=(1, 1), bias=False)
```

```
        (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      )
    )
  )
  (attn_block1): SpatialAttentionEncoder(
    (layer_norm1): LayerNorm((512,), eps=1e-05,
elementwise_affine=True)
    (mha): MultiheadAttention(
      (out_proj): NonDynamicallyQuantizableLinear(in_features=512,
out_features=512, bias=True)
    )
    (layer_norm2): LayerNorm((512,), eps=1e-05,
elementwise_affine=True)
    (ffn): Sequential(
      (0): Linear(in_features=512, out_features=2048, bias=True)
      (1): GELU(approximate='none')
      (2): Dropout(p=0.2, inplace=False)
      (3): Linear(in_features=2048, out_features=512, bias=True)
      (4): Dropout(p=0.2, inplace=False)
    )
  )
  (attn_block2): SpatialAttentionEncoder(
    (layer_norm1): LayerNorm((512,), eps=1e-05,
elementwise_affine=True)
    (mha): MultiheadAttention(
      (out_proj): NonDynamicallyQuantizableLinear(in_features=512,
out_features=512, bias=True)
    )
    (layer_norm2): LayerNorm((512,), eps=1e-05,
elementwise_affine=True)
    (ffn): Sequential(
      (0): Linear(in_features=512, out_features=2048, bias=True)
      (1): GELU(approximate='none')
      (2): Dropout(p=0.2, inplace=False)
      (3): Linear(in_features=2048, out_features=512, bias=True)
      (4): Dropout(p=0.2, inplace=False)
    )
  )
  (classifier): Sequential(
    (0): LayerNorm((512,), eps=1e-05, elementwise_affine=True)
    (1): Linear(in_features=512, out_features=3, bias=True)
  )
)

criterion = nn.CrossEntropyLoss()

optimizer = optim.AdamW(custom_hybrid_model.parameters(), lr=2e-4,
weight_decay=1e-4)
```

```
TOTAL_EPOCHS = 15

scheduler = CosineAnnealingLR(optimizer, T_max=TOTAL_EPOCHS)

print("Initiating Custom Hybrid Model Training...")
print("=" * 65)

for epoch in range(1, TOTAL_EPOCHS + 1):
    custom_hybrid_model.train()

    epoch_train_loss = 0.0
    train_correct_preds = 0
    total_samples = 0

    for batch_imgs, batch_labels in custom_train_loader:
        batch_imgs, batch_labels = batch_imgs.to(device),
batch_labels.to(device)

        optimizer.zero_grad()

        class_logits = custom_hybrid_model(batch_imgs)
        batch_loss = criterion(class_logits, batch_labels)

        batch_loss.backward()
        optimizer.step()

        epoch_train_loss += batch_loss.item() * batch_imgs.size(0)
        train_correct_preds += (class_logits.argmax(dim=1) ==
batch_labels).sum().item()
        total_samples += batch_imgs.size(0)

    scheduler.step()

    avg_epoch_loss = epoch_train_loss / total_samples
    epoch_accuracy = train_correct_preds / total_samples
    current_lr = scheduler.get_last_lr()[0]

    print(f'Epoch [{epoch:02d}/{TOTAL_EPOCHS}] | LR: {current_lr:.6f}
| Train Loss: {avg_epoch_loss:.4f} | Train Acc: {epoch_accuracy:.4f}')

print("=" * 65)
print("Training Complete!")
```

```
Initiating Custom Hybrid Model Training...
=================================================================
Epoch [01/15] | LR: 0.000198 | Train Loss: 0.7341 | Train Acc: 0.6882
Epoch [02/15] | LR: 0.000191 | Train Loss: 0.4808 | Train Acc: 0.8010
Epoch [03/15] | LR: 0.000181 | Train Loss: 0.3616 | Train Acc: 0.8601
Epoch [04/15] | LR: 0.000167 | Train Loss: 0.2991 | Train Acc: 0.8865
Epoch [05/15] | LR: 0.000150 | Train Loss: 0.2428 | Train Acc: 0.9126
Epoch [06/15] | LR: 0.000131 | Train Loss: 0.2223 | Train Acc: 0.9211
```

```
Epoch [07/15] | LR: 0.000110 | Train Loss: 0.1529 | Train Acc: 0.9493
Epoch [08/15] | LR: 0.000090 | Train Loss: 0.1113 | Train Acc: 0.9636
Epoch [09/15] | LR: 0.000069 | Train Loss: 0.0748 | Train Acc: 0.9747
Epoch [10/15] | LR: 0.000050 | Train Loss: 0.0468 | Train Acc: 0.9850
Epoch [11/15] | LR: 0.000033 | Train Loss: 0.0428 | Train Acc: 0.9855
Epoch [12/15] | LR: 0.000019 | Train Loss: 0.0239 | Train Acc: 0.9921
Epoch [13/15] | LR: 0.000009 | Train Loss: 0.0138 | Train Acc: 0.9963
Epoch [14/15] | LR: 0.000002 | Train Loss: 0.0119 | Train Acc: 0.9963
Epoch [15/15] | LR: 0.000000 | Train Loss: 0.0079 | Train Acc: 0.9979
==================================================================
Training Complete!
```

Evaluation

```
custom_hybrid_model.eval()

hybrid_predictions = []
hybrid_ground_truth = []

print("Evaluating Custom Hybrid Model on Validation Set...")

with torch.no_grad():
    for val_imgs, val_labels in custom_val_loader:
        val_imgs, val_labels = val_imgs.to(device),
val_labels.to(device)

        val_logits = custom_hybrid_model(val_imgs)
        predicted_classes = val_logits.argmax(dim=1)

        hybrid_predictions.extend(predicted_classes.cpu().numpy())
        hybrid_ground_truth.extend(val_labels.cpu().numpy())

hybrid_accuracy = accuracy_score(hybrid_ground_truth,
hybrid_predictions)

hybrid_prec_macro, hybrid_rec_macro, hybrid_f1_macro, _ =
precision_recall_fscore_support(
    hybrid_ground_truth, hybrid_predictions, average='macro',
zero_division=0
)

_, _, f1_per_class, _ = precision_recall_fscore_support(
    hybrid_ground_truth, hybrid_predictions, average=None,
zero_division=0
)

print("\n" + "="*50)
print(" PROPOSED HYBRID MODEL EVALUATION METRICS")
print("="*50)
print(f'Overall Accuracy         : {hybrid_accuracy:.4f}')
```

```
print(f'Macro Precision          : {hybrid_prec_macro:.4f}')
print(f'Macro Recall (mAR)       : {hybrid_rec_macro:.4f}')
print(f'Macro F1-Score           : {hybrid_f1_macro:.4f}')
print("-" * 50)

print("Per-Class F1-Scores:")
class_names = ['Abiotic', 'Insect', 'Disease']
for i, class_name in enumerate(class_names):
    score = f1_per_class[i] if i < len(f1_per_class) else 0.0
    print(f'  - {class_name:<10} : {score:.4f}')
print("="*50)
```

```
Evaluating Custom Hybrid Model on Validation Set...

==================================================
 PROPOSED HYBRID MODEL EVALUATION METRICS
==================================================
Overall Accuracy         : 0.8366
Macro Precision          : 0.8437
Macro Recall (mAR)       : 0.8411
Macro F1-Score           : 0.8422
--------------------------------------------------
Per-Class F1-Scores:
  - Abiotic    : 0.8642
  - Insect     : 0.8369
  - Disease    : 0.8254
==================================================
```

```
print(f'\n{"="*70}')
print(f'{"FINAL PERFORMANCE COMPARISON":^70}')
print(f'{"="*70}')
print(f'{"Evaluation Metric":<25} | {"YOLOv8s (Baseline)":>18} |
{"Proposed Hybrid Model":>20}')
print(f'{"-"*70}')
print(f'{"mAP@50":<25} | {baseline_mAP50:>18.4f} | {"N/A":>20}')
print(f'{"mAP@50-95 (Strict IoU)":<25} | {baseline_mAP50_95:>18.4f} |
{"N/A":>20}')
print(f'{"Mean Precision":<25} | {baseline_precision:>18.4f} |
{hybrid_prec_macro:>20.4f}')
print(f'{"Mean Average Recall (mAR)":<25} | {baseline_recall:>18.4f} |
{hybrid_rec_macro:>20.4f}')
print(f'{"Overall F1-Score":<25} | {baseline_f1:>18.4f} |
{hybrid_f1_macro:>20.4f}')
print(f'{"Overall Accuracy":<25} | {"N/A":>18} |
{hybrid_accuracy:>20.4f}')
print(f'{"="*70}')
```

```
======================================================================
                      FINAL PERFORMANCE COMPARISON
```

Evaluation Metric	YOLOv8s (Baseline)	Proposed Hybrid Model
mAP@50	0.3551	N/A
mAP@50-95 (Strict IoU)	0.1299	N/A
Mean Precision	0.4401	0.8437
Mean Average Recall (mAR)	0.3822	0.8411
Overall F1-Score	0.4091	0.8422
Overall Accuracy	N/A	0.8366